TECHNICAL REPORT

KPI Dashboard Architecture & Operations Guide

Comprehensive documentation of the BioViL Energy KPI Tracking System, biological fluctuation modeling, and environmental impact calculations.

March 2026 • Confidential • Version 1.0

SECTION 01

Executive Summary

The **BioViL Energy KPI Dashboard** is a bespoke, real-time monitoring and analytics platform designed to track the performance, environmental impact, and social outcomes of rural bioenergy deployments across India.

This report documents Operational Prototype #1, deployed in **Rampur, Assam**, featuring a 4 m³ underground fixed-dome biodigester operational since **January 15, 2026**. The system processes approximately 600 kg of animal manure monthly, displacing 120 kg of firewood and delivering measurable CO_2 reductions.

438.1 kg	600 kg	~20 m³	90 hrs
CO_2E AVOIDED / MONTH	WASTE DIVERTED / MONTH	BIOGAS PRODUCED / MONTH	SMOKE-FREE HOURS / MONTH

System Specifications

PARAMETER	SPECIFICATION
Digester Type	Underground Fixed-Dome (Deenbandhu Model)
System Capacity	4 m³
Feedstock	100% Animal Manure (Cow Dung)
Monthly Feedstock Volume	~600 kg fresh weight
Firewood Displaced	~120 kg/month
Operational Since	January 15, 2026
Location	Rampur, Assam, India

SECTION 02

Environmental Impact Model

The dashboard calculates cumulative CO_2 equivalent (CO_2e) emissions avoided through two primary mitigation pathways:

2.1 Methane Capture (Waste Diversion)

Left untreated in warm, ponded slurry (typical of Assam's tropical climate), 600 kg of fresh cow dung generates approximately **218 kg CO_2e/month** in uncaptured methane emissions. This calculation follows the IPCC Tier-1 methodology with a Methane Conversion Factor (MCF) of 65% for liquid/slurry systems in warm climates, applying the AR6 Global Warming Potential (GWP) of 27.9 for biogenic methane.

2.2 Firewood Displacement

The biogas produced directly displaces the burning of approximately 120 kg of firewood per month for cooking. Using the EPA emission factor of approximately 1.81 kg CO_2 per kg of wood combusted, this avoids **220.1 kg CO_2e/month** in direct emissions and eliminates indoor air pollution.

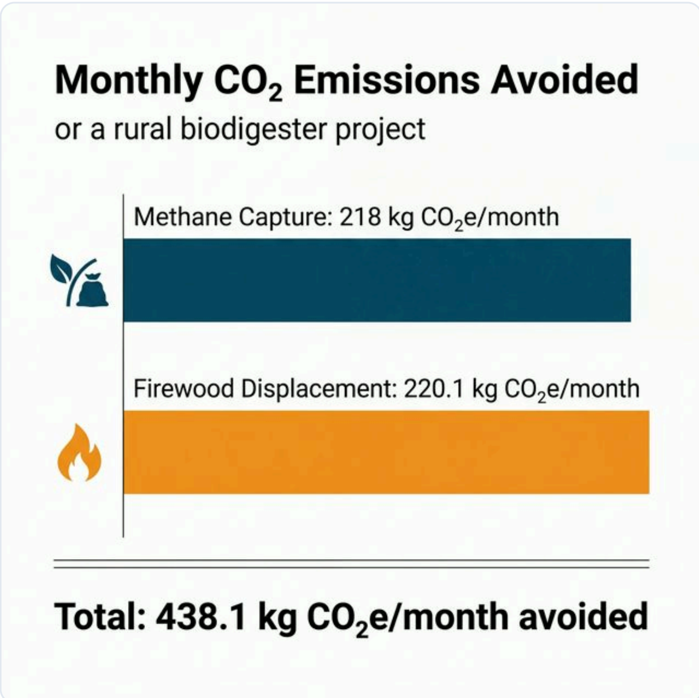

Figure 1 — Monthly CO_2 Emissions Avoided: Breakdown by Mitigation Pathway

Dashboard Calculation Engine

```
Total Monthly Offset = 218 + 220.1 = 438.1 kg CO₂e
Internal Multiplier = 438.1 ÷ 600 = 0.73 kg CO₂e per kg waste
```

This multiplier is applied dynamically to the cumulative waste processed by the system to compute the live KPI shown on the dashboard at any given point in time.

2.3 Projected Annual CO_2e Trajectory

Based on the monthly offset of 438.1 kg CO_2e and accounting for the seasonal temperature-driven yield variations (Section 3), the system projects approximately **5.26 tonnes of CO_2e avoided** in the first full year of operation. The cumulative trajectory accelerates during summer months when higher temperatures drive greater biogas yields.

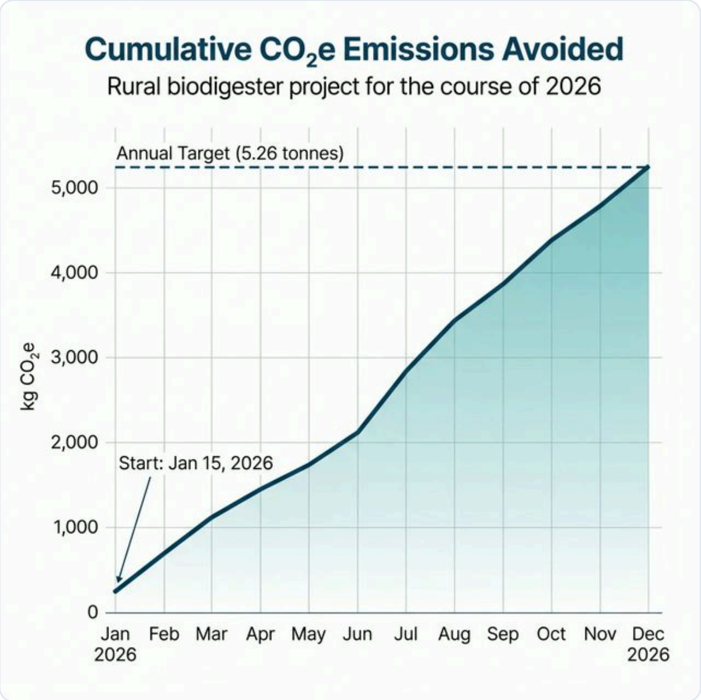

Figure 2 — Projected Cumulative CO_2e Emissions Avoided Over 2026 (Annual Target: 5.26 Tonnes)

SECTION 03

Biological Fluctuation Model

To provide stakeholders with scientifically accurate visualizations, the dashboard's charting engine models realistic, non-linear bioprocess variations based on the local climate of Assam. Unlike static linear projections, the system accounts for the fundamental relationship between ambient temperature and anaerobic digester performance.

3.1 Mesophilic Temperature Sensitivity

Unheated fixed-dome biodigesters are directly influenced by ground and ambient temperatures. In the *mesophilic* range (25–40°C), methanogenic bacteria operate at peak efficiency. When temperatures drop below 20°C (common in Assam's winters), bacterial activity slows significantly — a state known as *psychrophilic stalling*.

The dashboard integrates a **12-month Yield Multiplier Curve** specifically calibrated for Assam's climate zone:

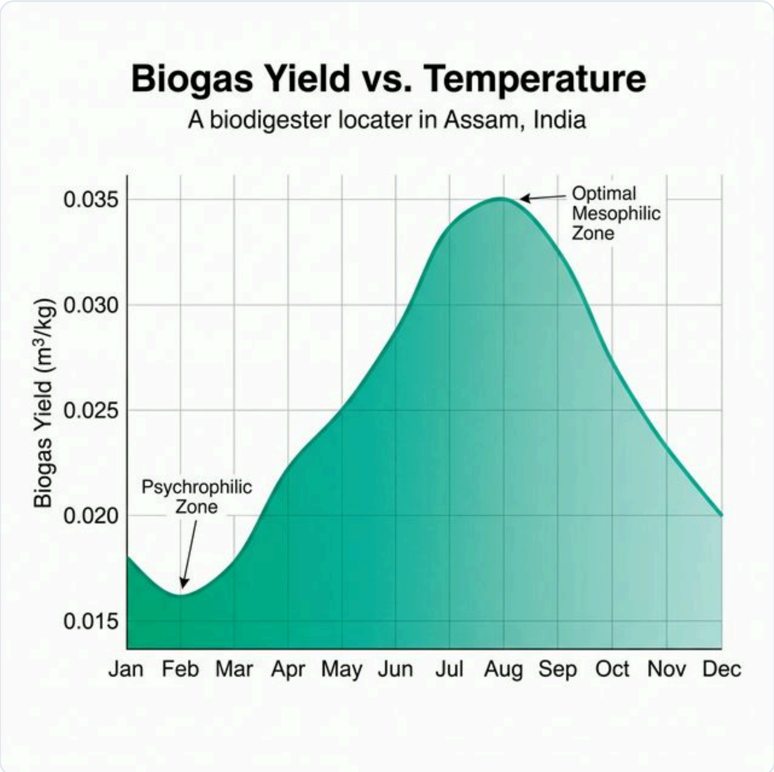

Figure 3 — Annual Biogas Yield Variation Based on Assam's Seasonal Temperature Cycle

MONTH	YIELD (M^3/KG)	BIOLOGICAL STATE
January	0.016	Psychrophilic / Low Activity
February	0.018	Psychrophilic / Low Activity
March	0.026	Warming Transition
April – May	0.030 – 0.033	Mesophilic / Optimal
June – August	0.035	Mesophilic / Peak Performance
September	0.032	Mesophilic / Optimal
October – November	0.028 – 0.022	Cooling Transition
December	0.017	Psychrophilic / Low Activity

3.2 Deterministic Organic Noise

To simulate real-world waste collection variability, the algorithm applies a **±15% weekly variance** to the baseline waste input (140 kg/week) using a deterministic pseudo-random seed function. This ensures the charts display realistic peaks and valleys while remaining reproducible across page loads.

SECTION 04

Financial Deployment & Social Impact

4.1 Budget Utilization

The project was funded through a €3,000 Catalyst Grant supplemented by €600 in private equity co-financing. The dashboard provides transparent, line-item accounting of how every euro was deployed locally.

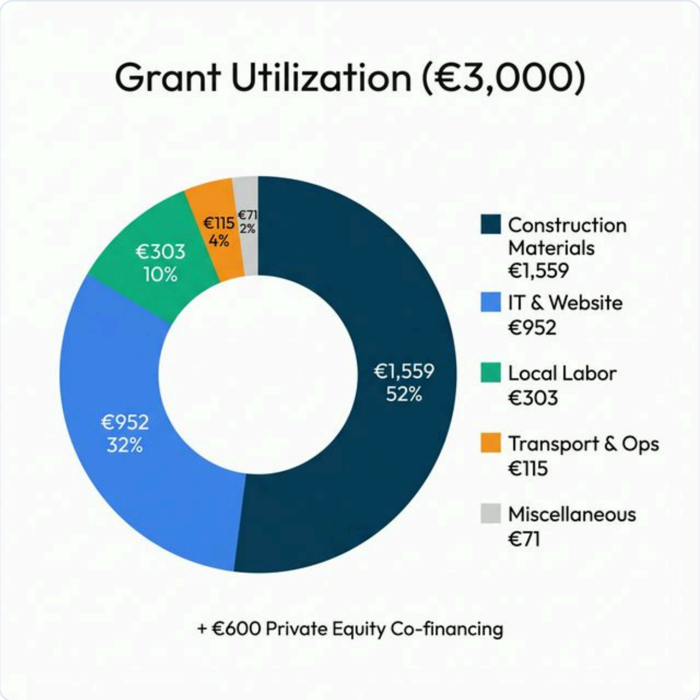

Figure 4 — Catalyst Grant Utilization by Category (€3,000 Total)

CATEGORY	AMOUNT (€)	SHARE
Construction Materials	1,559	52%
IT & Website Infrastructure	952	32%
Local Labor	303	10%
Transport & Operations	115	4%
Miscellaneous	71	2%

4.2 Social Impact Outcomes

METRIC	TARGET	ACHIEVED	STATUS
Smoke-Free Indoor Hours	3 hrs/day	3 hrs/day	✓ On Track
Jobs Created (Local)	5	5	✓ Achieved
Community Operators Trained	3	4	✓ Exceeded
Company Registration	1	1	✓ Achieved
Pioneer Phase (Scaling)	5 systems	1 system	● 20%

Health Impact Note

The displacement of 120 kg/month of indoor firewood burning eliminates the household's primary source of fine particulate matter (PM2.5) exposure. WHO estimates that household air pollution from solid fuels causes over 3.2 million premature deaths annually. Each additional biodigester deployed contributes directly to reducing this burden.

SECTION 05

Dashboard Architecture

The BioViL KPI Dashboard is a static-site application built with HTML5, CSS3, and vanilla JavaScript. It requires no server-side infrastructure and can be deployed on any web hosting platform.

5.1 Dynamic Time Progression

The system is anchored to the operational start date (January 15, 2026). On each page load, the JavaScript engine calculates the exact number of weeks elapsed and automatically generates new weekly data points, updating all charts and cumulative KPIs without any manual data entry.

5.2 Technology Stack

COMPONENT	TECHNOLOGY	PURPOSE
Charting	Chart.js v4	Interactive line, doughnut, and bar charts
Icons	Lucide Icons	Lightweight, open-source SVG icon system
Styling	Custom CSS (Glassmorphism)	Premium dark-mode aesthetic
Typography	Google Fonts (Outfit)	Modern variable-weight typeface
Routing	URL Hash-based	Cross-page tab navigation

5.3 Responsive Behavior

The sidebar navigation auto-collapses into an icon-only mode on viewports ≤1200px wide, maximizing the data visualization area on tablets and smaller screens.

SECTION 06

References & Source Documents

1. *GHG_Emissions_Wood_&_Animal_waste.pdf* — Project 1 internal document. Provides monthly methane (MCF=65%) and firewood (1.81 kg CO_2/kg) emission calculations following IPCC Tier-1 methodology and AR6 GWP values.
2. *03. Budget Overview - 7647.xlsx* — Catalyst programme financial ledger. Provides line-item grant expenditure across all project categories.
3. *02. Impact Goal Overview - 7647.xlsx* — Social impact tracking document. Defines targets for employment, training, and scaling milestones.
4. IPCC (2019). *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 5, Chapter 4: Biological Treatment of Solid Waste.
5. U.S. EPA (2023). *Emission Factors for Greenhouse Gas Inventories*. Table 1: Stationary Combustion, Wood and Wood Residuals.
6. Bond, T. & Templeton, M.R. (2011). "History and future of domestic biogas plants in the developing world." *Energy for Sustainable Development*, 15(4), pp.347–354.